ACADEMIC TECHNICAL DOCUMENTATION • MINOR PROJECT

MONO TIME — Implementation Guide & Engineering Architecture

A Comprehensive Step-by-Step Breakdown of How Every Subsystem Was Engineered From Start to Finish

Domain: Full-Stack Web Development & Autonomous AI | **Evaluation:** College Minor Project Defense | **Status:** Production Ready

1. Project Objective & Core Problem Statement

Traditional luxury e-commerce platforms suffer from flat 2D product photos that fail to convey physical craftsmanship, disconnected chatbots that hallucinate inventory and flood screens with intrusive product cards, and mock checkouts that never reach a persistent database. Mono Time was built to solve these three fundamental challenges through an integrated 3D procedural engine, an autonomous Gemini AI concierge, and an end-to-end relational order reflection pipeline.

2. Step-by-Step Implementation Roadmap (Phases 1 to 12)

• **Phase 1: Real-Time Multi-Attribute Catalog & Instant Search:** Built client-side search in store.tsx indexing watch title, brand, movement, dial color, and description with zero keystroke latency. Clicking 'All Watches' clears category filters to display the complete 15-piece catalog.

• **Phase 2: Dynamic Category Decoupling & Flagship Hero Switching:** Decoupled the hero section so that choosing Men's Collection automatically sets the featured showcase to the Men's Flagship (Titan Edge Minimalist Slate Steel), and choosing Women's Collection switches to the Women's Haute Horology Flagship (Titan Raga Flora Crystal Bangle).

• **Phase 3: Haute Horology Emerald & Obsidian Design System:** Crafted an exclusive colorway inspired by Swiss luxury houses: Obsidian Noir (#09090b), Emerald Malachite (#10b981), and Platinum Silver (#d1d5db). Integrated radial spotlight card reveal hovers, dial shimmer light sweeps, and card lift elevations.

• **Phase 4: Procedural 3D Classic Watch Engineering (Three.js WebGL):** Engineered Watch3DBackground.tsx using procedural trigonometry: a 72-facet coin-edge fluted gold bezel, 1024x1024 high-resolution radial guilloché canvas dial with 120 lathe rays, 8Hz oscillating tourbillon escapement, 5-link Jubilee bracelet, and cursor/touch parallax kinematics.

• **Phase 5: Google Gemini AI Multimodal Concierge & UI De-Cluttering:** Connected the chat widget to Google Gemini 1.5 Flash. Taught the AI complete specifications of all 15 watches, store policies, and Indian support desk contacts. Removed bulky product preview cards from chat bubbles so concierge responses render in spacious, clean bubbles without forcing users to scroll upward.

• **Phase 6: Catalog Lineage Expansion (15 Curated References):** Expanded inventory across iconic brands: Titan (Grandmaster Skeleton, Edge, Raga Emerald Ceramic, Raga Flora), Casio (Vintage A168 Retro Chrome, Edifice Chrono, Sheen), Timex (Marlin Emerald Malachite, Transcend Gold Mesh, Waterbury), Fossil (Carlie Mini Diamond, Jacqueline, Neutra), and Aethelgard. Integrated high-res studio photography.

• **Phase 7: Checkout & Voucher Rebranding (MONOTIME10):** Rebranded checkout gateway to 'Mono Time Secure Gateway' and standardized privilege discount code to 'MONOTIME10' (-10% instant discount). Added in-modal voucher application card and localized telephone contacts to Indian format: 1800-266-8463 (Toll Free) and +91 98200 66684.

• **Phase 8: Full Frontend-to-Backend Order Reflection Pipeline:** Engineered POST /api/checkout/confirm in checkout.py to parse complete serialized cart payloads (line items, customer name, email, city, amount paid, voucher used) and persist records into SQLite checkout_orders table. Generated real order tracking IDs (ORD-YYYYMMDD-XXXXXX).

• **Phase 9: Live Merchant ROI & Client Acquisition Analytics Dashboard:** Constructed /merchant dashboard polling directly from GET /api/checkout/orders. Store administrators can inspect real-time customer acquisitions, voucher redemptions,

exit-intent conversion rates, and gross revenue.

• **Phase 10: Role-Based Access Control Architecture (Customer vs Admin Mode):** Segregated customer shopping experience from administrative analytics. In Customer Mode, all internal ROI links and PRD banners are hidden, replaced with 'The Mono Time Horology Atelier'. In Admin Mode, an illuminated toolbar unlocks merchant analytics, ROI metrics, and PRD evaluation tools.

• **Phase 11: Smartphone & Touchscreen Mobile Optimization:** Configured responsive viewport scaling, mobile-adaptive 3D camera distance (z=32), touchmove parallax physics, horizontal-scroll category filter pills, edge-to-edge chat drawer (inset-x-2 bottom-2), and compact touch-friendly checkout modal.

• **Phase 12: Production Build Validation & 100% Test Suite Verification:** Executed Pytest test suite: 8/8 tests passed (100%). Fixed TypeScript type definitions (added 'concierge' to agentType union) and verified Next.js production build: npm run build completed with 0 errors across all routes.

3. Technical Verification & Test Pass Rate Summary

Test Function	Module Under Test	Verification Target	Result
test_health_and_root	Backend API Gateway	Verifies root status 200 OK and health endpoint	**PASSED**
test_get_products	Catalog Router	Verifies all seeded luxury timepieces returned	**PASSED**
test_product_search	Search Router	Validates multi-attribute search across dial and brand	**PASSED**
test_agent_chat_sizing_intent	Gemini Agent Graph	Validates wrist sizing advice and stylist routing	**PASSED**
test_agent_chat_rag_intent	RAG Agent Graph	Ensures grounded factual responses without hallucination	**PASSED**
test_exit_intent_observer	Observer Router	Verifies dynamic exit-intent trigger and MONOTIME10 offer	**PASSED**
test_merchant_analytics	Merchant Router	Validates session tracking, pitch counts & conversion stats	**PASSED**
test_mock_checkout	Checkout Router	Asserts order persistence in SQLite and retrieval parity	**PASSED**

4. Recommended Viva / Evaluation Demonstration Flow

• **Step 1: Visual & 3D Immersion:** Show the procedural 3D watch in Three.js with fluted gold bezel, openwork tourbillon movement, and cursor parallax tracking. Highlight the Emerald & Obsidian luxury aesthetic.

• **Step 2: Dynamic Catalog & Search:** Type 'Automatic' or 'Titan' in search bar to show instant filtering. Click 'Men's Collection' and 'Women's Collection' to show dynamic flagship banner switching.

• **Step 3: Google Gemini AI Concierge:** Ask a question: 'Which watch has an automatic skeleton dial?' Show Gemini answering concisely with zero hallucinations, in readable bubbles without obstructive cards.

• **Step 4: End-to-End Checkout:** Add Titan Grandmaster Skeleton to cart, apply voucher MONOTIME10 (-10%), enter customer details, and confirm payment. Note the generated real Order ID (ORD-XXXX).

• **Step 5: Backend Order Reflection:** Switch to Admin Mode and open /merchant. Show the professor the newly placed order appearing live in the database table with customer email, exact amount paid, and purchased items!